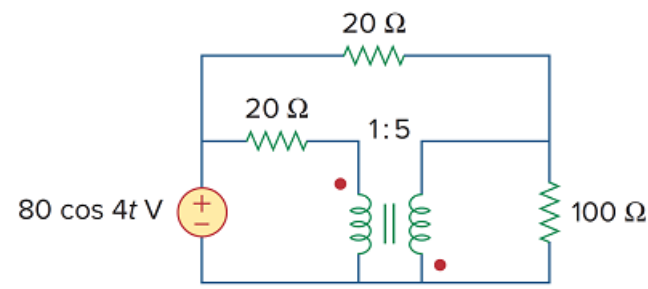


Problem

Determine the average power absorbed by each resistor in the circuit of Fig. 13.123.

Figure 13.123



Step-by-step solution

Step 1 of 8

Refer to Figure 13.123 in the textbook.

Convert the time domain input voltage to phasor form.

$$80 \cos 4t \text{ V} = 80 \angle 0^\circ \text{ V}$$

Draw the frequency domain with the loop currents.

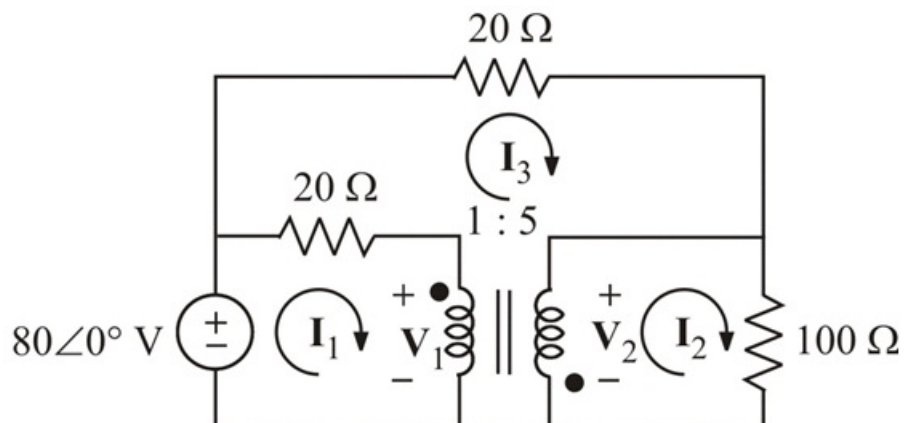


Figure 1

Step 2 of 8

Calculate the relation between primary and secondary currents.

$$\frac{I_2}{I_1} = -\frac{N_1}{N_2}$$

$$\frac{I_2}{I_1} = -\frac{1}{5}$$

$$I_2 = -0.2I_1$$

Apply Kirchhoff's voltage law to loop 2.

$$V_2 = 100I_2$$

Substitute $-0.2I_1$ for I_2 .

$$\begin{aligned} V_2 &= 100(-0.2I_1) \\ &= -20I_1 \end{aligned}$$

Step 3 of 8

Calculate the relation between primary and secondary voltages.

$$\frac{V_2}{V_1} = -\frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = -\frac{5}{1}$$

$$V_1 = -0.2V_2$$

Substitute $-20I_1$ for V_2 .

$$\begin{aligned} V_1 &= -0.2(-20I_1) \\ &= 4I_1 \end{aligned}$$

Step 4 of 8

Apply Kirchhoff's voltage law to loop 1.

$$20(I_1 - I_3) + V_1 = 80$$

$$20I_1 - 20I_3 + V_1 = 80$$

Substitute $4I_1$ for V_1 .

$$20I_1 - 20I_3 + 4I_1 = 80$$

$$24I_1 - 20I_3 = 80$$

$$20I_3 = 24I_1 - 80$$

$$I_3 = 1.2I_1 - 4$$

Step 5 of 8

Apply Kirchhoff's voltage law to loop 3

$$-V_1 + 20(I_3 - I_1) + 20I_3 + V_2 = 0$$

$$-V_1 + 20I_3 - 20I_1 + 20I_3 + V_2 = 0$$

$$-20I_1 + 40I_3 - V_1 + V_2 = 0$$

Substitute $4I_1$ for V_1 , $-20I_1$ for V_2 and $1.2I_1 - 4$ for I_3 .

$$-20I_1 + 40(1.2I_1 - 4) - 4I_1 - 20I_1 = 0$$

$$-20I_1 + 48I_1 - 160 - 4I_1 - 20I_1 = 0$$

$$4I_1 = 160$$

$$I_1 = 40 \text{ A}$$

Step 6 of 8

Calculate the current, I_2 .

$$I_2 = -0.2I_1$$

Substitute 40 A for I_1 .

$$\begin{aligned} I_2 &= -0.2(40 \text{ A}) \\ &= -8 \text{ A} \end{aligned}$$

Calculate the current, I_3 .

$$I_3 = 1.2I_1 - 4$$

Substitute 40 A for I_1 .

$$\begin{aligned} I_3 &= 1.2(40) - 4 \\ &= 44 \text{ A} \end{aligned}$$

Step 7 of 8

Calculate the average power absorbed by the 20Ω resistor in loop 1.

$$P_{20\Omega} = \frac{1}{2}(I_1 - I_3)^2(20)$$

Substitute 40 A for I_1 and 44 A for I_3 .

$$\begin{aligned} P_{20\Omega} &= \frac{1}{2}(40 - 44)^2(20) \\ &= (4)^2(10) \\ &= 160 \text{ W} \end{aligned}$$

Therefore, the average power absorbed by 20Ω resistor in loop 1 is 160 W.

Step 8 of 8

Calculate the average power absorbed by 100Ω resistor.

$$P_{100\Omega} = \frac{1}{2}(I_2^2)(100)$$

Substitute -8 A for I_2 .

$$\begin{aligned} P_{100\Omega} &= \frac{1}{2}(-8)^2(100) \\ &= (64)(50) \\ &= 3200 \text{ W} \end{aligned}$$

Therefore, the power absorbed by 100Ω resistor is 3200 W.

Calculate the average power absorbed by another 20Ω resistor in loop 3.

$$P_{20\Omega} = \frac{1}{2}(I_3)^2(20)$$

Substitute 44 A for I_3 .

$$\begin{aligned} P_{20\Omega} &= \frac{1}{2}(44)^2(20) \\ &= (44)^2(10) \\ &= 19360 \text{ W} \end{aligned}$$

Therefore, the average power absorbed by 20Ω resistor in loop 3 is 19360 W.